Problem - Solution Fit

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Problem - Solution Fit Canvas:

Floods cause severe damage to lives, property, and the environment, making accurate early prediction essential for effective disaster management. This project develops a Machine Learning-based Flood Prediction System that analyzes historical weather and rainfall data to predict flood risk using classification algorithms. The best-performing model is integrated into a Flask web application to provide instant and reliable flood predictions. The system supports early warning, informed decision-making, and efficient disaster preparedness through scalable cloud deployment.

Purpose:

- Develop an intelligent Machine Learning system to accurately predict flood risk using historical weather and rainfall data.
- Compare multiple classification algorithms and select the best-performing model based on evaluation metrics.
- Perform data preprocessing, visualization, and feature engineering to improve prediction accuracy.
- Build a user-friendly Flask web application that provides instant flood prediction results based on user inputs.
- Support disaster management authorities by enabling early warning, evacuation planning, and efficient resource allocation.
- Deploy the solution on IBM Cloud for scalable, accessible, and real-time flood risk monitoring.
- These points cover the complete lifecycle of your project, from data collection and model development to deployment and real-world application.

Template:

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) — CS

Who is your customer?

- Disaster Management Authorities
- Government Agencies
- Meteorological Departments
- Emergency Response Teams
- Local Administration
- NGOs and Relief Organizations

6. CUSTOMER CONSTRAINTS — CC

What constraints prevent your customers from taking action or limit their choices of solutions? (i.e. limited forecasting accuracy, delayed warnings, lack of real-time monitoring, high manual effort, complex tools, limited accessibility, etc.)

- Inaccurate and delayed flood predictions
- Lack of real-time weather data analysis
- Limited access to advanced prediction tools
- Manual monitoring is time-consuming and error-prone
- Insufficient early warning leads to high damage and loss

5. AVAILABLE SOLUTIONS — AS

Which solutions are available to the customers when they face the problem or need to get the job done? What have they used in the past? What pros & cons do these solutions have? (i.e. traditional forecasting, manual methods, basic alerts, standalone software, etc.)

- Traditional weather forecasting
- Manual data analysis and reporting
- Basic flood alert systems (sirens, SMS)
- Standalone software with limited accuracy
- General mobile apps with low prediction capability

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which job-to-be-done (or problems) do your customers try to solve?

- Predict flood risk accurately and in advance
- Monitor rainfall and weather conditions
- Issue early warnings to at-risk communities
- Plan evacuations and resource allocation
- Reduce loss of life and property

9. PROBLEM ROOT CAUSE — RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
(i.e. root causes that trigger the problem)

- High variability in weather and rainfall patterns
- Inability to analyze large historical data manually
- Lack of intelligent prediction models
- Delayed information flow to decision makers
- Inadequate technology adoption in disaster management

7. BEHAVIOUR — BE

What does your customer do to address the problem and get the job done? (i.e. how do they currently try to solve it? Their habits, workarounds, existing efforts)

- Rely on manual reports and past experience
- Monitor weather updates from multiple sources
- Use spreadsheets and basic tools for analysis
- Communicate through calls and messages
- Take action after severe conditions are observed

Focus on J&P, tap into BE, understand RC

Identify TR & EM

3. TRIGGERS — TR

What triggers customers to act? (i.e. noticing heavy rain, rising water levels, alerts from IMD, local reports, previous flood experiences, etc.)

- Heavy rainfall and cloud build-up
- River water level rising
- Weather department alerts
- Past flood experiences in the region
- Local reports and visual observations

4. EMOTIONS: BEFORE / AFTER — EM

How do customers feel when they face a problem or a job and afterwards? (i.e. anxiety, fear, uncertainty before; relief, confidence, safety after)

- Before: Anxiety, uncertainty, fear, stress
- After: Relief, confidence, preparedness, safety

10. YOUR SOLUTION — SL

If you are working on an existing business, write down your current solution first.
If in the new venture, define how your RPGS (Rising Waters Flood Prediction System) is a solution.
(i.e. how our ML-based system analyzes weather data, predicts flood risk, provides early warnings, helps in decision making, and supports disaster management.)

- Machine Learning models predict flood risk using historical weather data (rainfall, cloud visibility, seasonal patterns, etc.)
- Multiple algorithms (DT, RF, KNN, XGBoost) ensure high accuracy
- Flask web application provides instant predictions
- Early warnings help in evacuation and resource planning
- Deployed on IBM Cloud for real-time, scalable access

8. CHANNELS of BEHAVIOUR — CH

8.1 ONLINE

What kind of actions do customers take online? Extract online channels from #7

- Access web application
- Check flood risk predictions online
- Receive alerts and notifications
- Share reports through digital platforms

8.2 OFFLINE

What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

- Field visits and inspections
- Community meetings and announcements
- Printed reports and notices
- Coordination through phone calls

Extract online & offline CH of BE

Project: Rising Waters – A Machine Learning Approach to Flood Prediction

Empowering Communities with Early Warnings, Accurate Predictions, and Smarter Disaster Management.